1. Uses lag = 2, standard 3D LSTM input shape (samples, timesteps, features), sequences created using a sliding window of past *lag* steps.

```
✅ LSTM Input shape: (2878, 2, 31), Target shape: (2878,)
📊 Train samples: 2302, Test samples: 576
/usr/local/lib/python3.11/dist-packages/keras/src/layers/rnn/rnn.py:200: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer usin
  super().__init__(**kwargs)
Model: "sequential_4"
┌─────────────────────────┬──────────────────┬──────────┐
│ Layer (type)            │ Output Shape     │  Param # │
├─────────────────────────┼──────────────────┼──────────┤
│ lstm_4 (LSTM)           │ (None, 64)       │   24,576 │
├─────────────────────────┼──────────────────┼──────────┤
│ dropout_2 (Dropout)     │ (None, 64)       │        0 │
├─────────────────────────┼──────────────────┼──────────┤
│ dense_6 (Dense)         │ (None, 32)       │    2,080 │
├─────────────────────────┼──────────────────┼──────────┤
│ dense_7 (Dense)         │ (None, 1)        │       33 │
└─────────────────────────┴──────────────────┴──────────┘
 Total params: 26,689 (104.25 KB)
 Trainable params: 26,689 (104.25 KB)
 Non-trainable params: 0 (0.00 B)
None
Epoch 1/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 4s 8ms/step - loss: 0.1580 - val_loss: 0.0209
Epoch 2/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0276 - val_loss: 0.0242
Epoch 3/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0270 - val_loss: 0.0184
Epoch 4/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0251 - val_loss: 0.0193
Epoch 5/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0244 - val_loss: 0.0186
Epoch 6/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0254 - val_loss: 0.0186
Epoch 7/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0239 - val_loss: 0.0188
Epoch 8/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0237 - val_loss: 0.0188
```

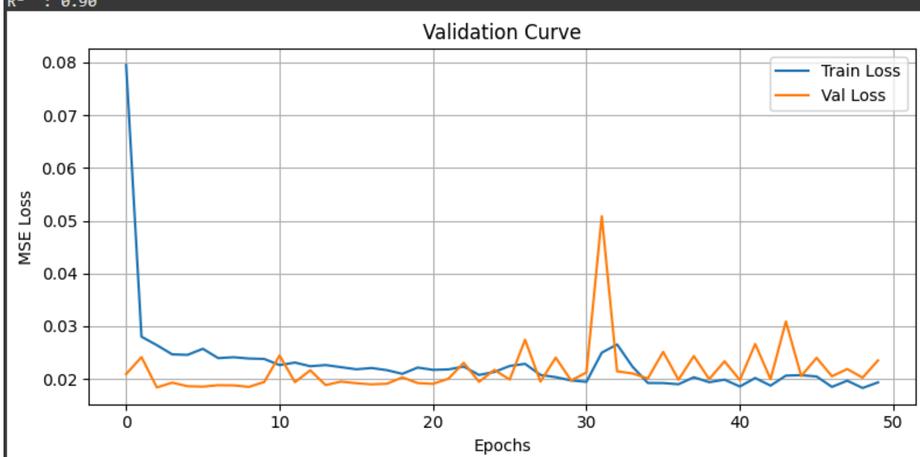

Epoch 8/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0237 - val_loss: 0.0188


```
Epoch 9/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0233 - val_loss: 0.0185
Epoch 10/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0244 - val_loss: 0.0195
Epoch 11/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0228 - val_loss: 0.0245
Epoch 12/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.0212 - val_loss: 0.0194
Epoch 13/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0228 - val_loss: 0.0217
Epoch 14/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0225 - val_loss: 0.0189
Epoch 15/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0228 - val_loss: 0.0195
Epoch 16/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0208 - val_loss: 0.0192
Epoch 17/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0226 - val_loss: 0.0190
Epoch 18/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0217 - val_loss: 0.0191
Epoch 19/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0217 - val_loss: 0.0203
Epoch 20/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0228 - val_loss: 0.0192
Epoch 21/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0219 - val_loss: 0.0191
Epoch 22/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0210 - val_loss: 0.0201
Epoch 23/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0231 - val_loss: 0.0231
Epoch 24/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0201 - val_loss: 0.0195
Epoch 25/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0207 - val_loss: 0.0217
```

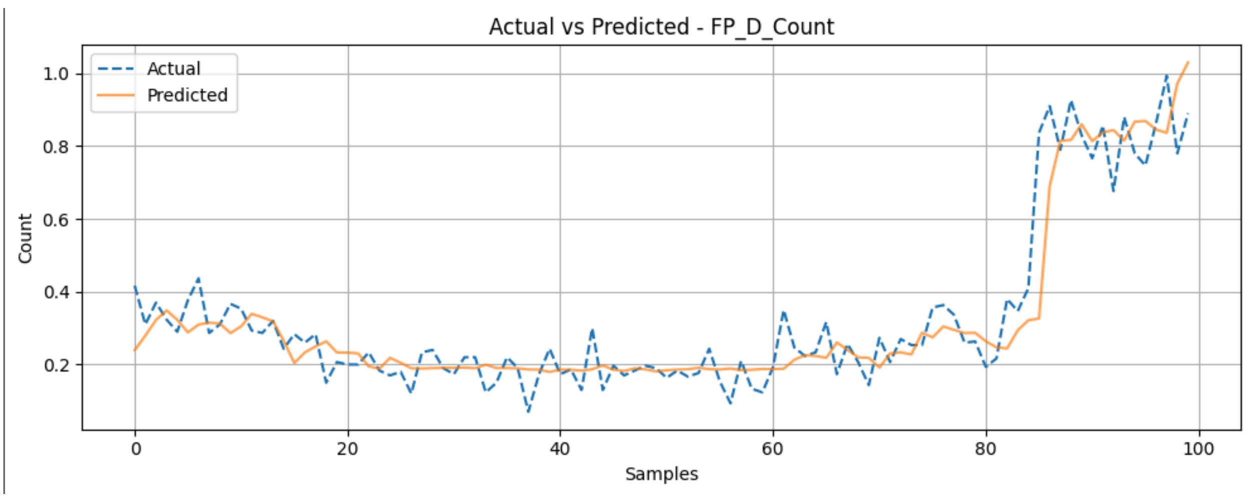
```
Epoch 26/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0234 - val_loss: 0.0199
Epoch 27/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0232 - val_loss: 0.0274
Epoch 28/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0205 - val_loss: 0.0195
Epoch 29/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0210 - val_loss: 0.0241
Epoch 30/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0204 - val_loss: 0.0197
Epoch 31/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0194 - val_loss: 0.0213
Epoch 32/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0248 - val_loss: 0.0508
Epoch 33/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0254 - val_loss: 0.0215
Epoch 34/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0256 - val_loss: 0.0211
Epoch 35/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0193 - val_loss: 0.0201
Epoch 36/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0199 - val_loss: 0.0251
Epoch 37/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0191 - val_loss: 0.0199
Epoch 38/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0211 - val_loss: 0.0244
Epoch 39/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0193 - val_loss: 0.0200
Epoch 40/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0203 - val_loss: 0.0234
Epoch 41/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0181 - val_loss: 0.0198
Epoch 42/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0209 - val_loss: 0.0266
Epoch 43/50
```

```
Epoch 43/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0185 - val_loss: 0.0200
Epoch 44/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0215 - val_loss: 0.0309
Epoch 45/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0200 - val_loss: 0.0206
Epoch 46/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0223 - val_loss: 0.0240
Epoch 47/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.0187 - val_loss: 0.0205
Epoch 48/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0203 - val_loss: 0.0219
Epoch 49/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0183 - val_loss: 0.0203
Epoch 50/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0201 - val_loss: 0.0235
18/18 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
```

```
📈 Final Evaluation:
MAE : 0.09
RMSE: 0.13
R²  : 0.90
```



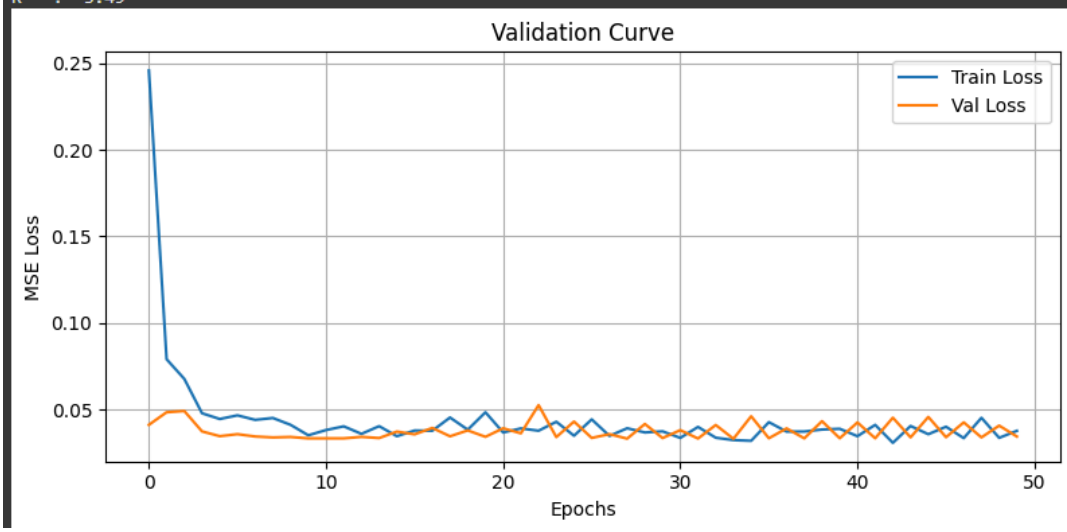
Actual vs Predicted - FP_D_Count

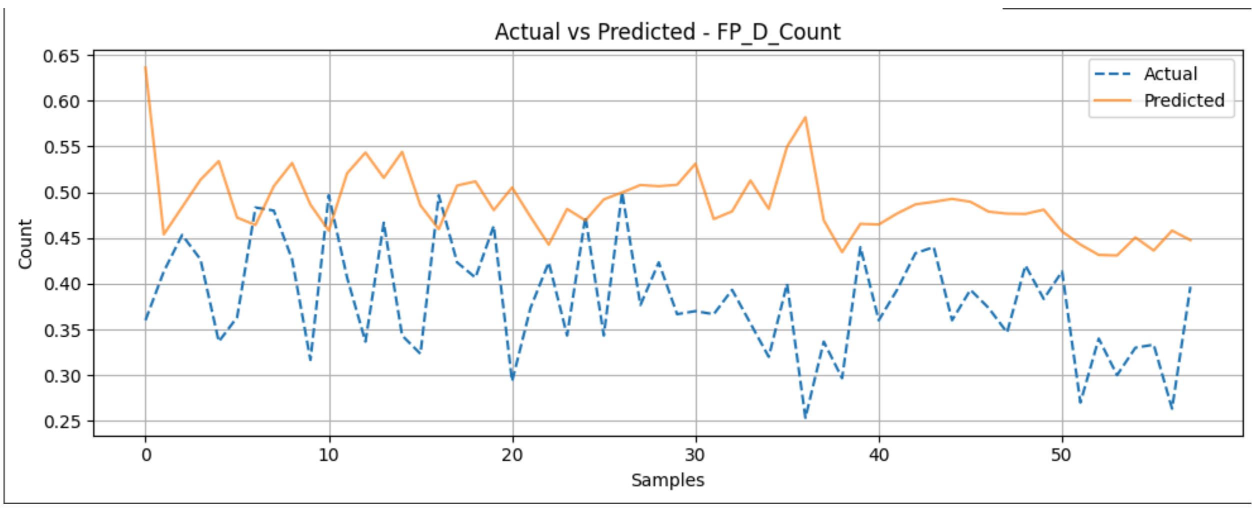


2. Using the past 2 time steps of all numeric features from a 5-minute downsampled traffic dataset.

```
Epoch 48/50
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 0.0288 - val_loss: 0.0340
Epoch 49/50
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.0227 - val_loss: 0.0408
Epoch 50/50
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 0.0241 - val_loss: 0.0345
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 198ms/step

📈 Final Evaluation:
MAE : 0.11
RMSE: 0.13
R²  : -3.49
```





3. Uses lag = 5 and explicitly flattens lagged features into 2D, then reshapes them back to 3D before LSTM (columns explicitly labeled as `col_t-1`, `col_t-2`, etc.).

```
✅ Lagged DataFrame shape: (2875, 155), Target shape: (2875,)
📊 Train samples: 2300, Test samples: 575

🔍 X_train.columns:
['Time_t-5', 'FP_A_Count_t-5', 'FP_A_QueuePct_t-5', 'FP_A_Speed_t-5', 'FP_B_Count_t-5', 'FP_B_QueuePct_t-5', 'FP_B_Speed_t-5', 'FP_C_Count_t-5', 'FP_C_QueuePct_t-5', 'FP_C_Speed_t-5', 'FP_D_

🔍 X_test.columns:
['Time_t-5', 'FP_A_Count_t-5', 'FP_A_QueuePct_t-5', 'FP_A_Speed_t-5', 'FP_B_Count_t-5', 'FP_B_QueuePct_t-5', 'FP_B_Speed_t-5', 'FP_C_Count_t-5', 'FP_C_QueuePct_t-5', 'FP_C_Speed_t-5', 'FP_D_
/usr/local/lib/python3.11/dist-packages/keras/src/layers/rnn/rnn.py:200: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using
  super().__init__(**kwargs)
Model: "sequential_7"
```

Layer (type)	Output Shape	Param #
lstm_7 (LSTM)	(None, 64)	24,576
dropout_5 (Dropout)	(None, 64)	0
dense_12 (Dense)	(None, 32)	2,080
dense_13 (Dense)	(None, 1)	33

```
 Total params: 26,689 (104.25 KB)
 Trainable params: 26,689 (104.25 KB)
 Non-trainable params: 0 (0.00 B)
None
Epoch 1/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 0.0838 - val_loss: 0.0277
```

```
Epoch 2/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0296 - val_loss: 0.0236
Epoch 3/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0267 - val_loss: 0.0251
Epoch 4/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0248 - val_loss: 0.0218
Epoch 5/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0253 - val_loss: 0.0228
Epoch 6/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0258 - val_loss: 0.0268
Epoch 7/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0237 - val_loss: 0.0234
Epoch 8/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0247 - val_loss: 0.0241
Epoch 9/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0243 - val_loss: 0.0205
Epoch 10/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0225 - val_loss: 0.0201
Epoch 11/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 0.0238 - val_loss: 0.0199
Epoch 12/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - loss: 0.0225 - val_loss: 0.0212
Epoch 13/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0227 - val_loss: 0.0215
Epoch 14/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0213 - val_loss: 0.0201
Epoch 15/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0227 - val_loss: 0.0200
```

```
Epoch 16/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0223 - val_loss: 0.0241
Epoch 17/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0216 - val_loss: 0.0197
Epoch 18/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0223 - val_loss: 0.0273
Epoch 19/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0203 - val_loss: 0.0215
Epoch 20/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0213 - val_loss: 0.0273
Epoch 21/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0202 - val_loss: 0.0214
Epoch 22/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0208 - val_loss: 0.0334
Epoch 23/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0197 - val_loss: 0.0210
Epoch 24/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0214 - val_loss: 0.0278
Epoch 25/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0197 - val_loss: 0.0250
Epoch 26/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0200 - val_loss: 0.0262
Epoch 27/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0203 - val_loss: 0.0266
Epoch 28/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0206 - val_loss: 0.0296
Epoch 29/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0202 - val_loss: 0.0356
Epoch 30/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 0.0184 - val_loss: 0.0239
Epoch 31/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0190 - val_loss: 0.0332
Epoch 32/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0191 - val_loss: 0.0269
Epoch 33/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0194 - val_loss: 0.0327
```

```
Epoch 34/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0185 - val_loss: 0.0205
Epoch 35/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0222 - val_loss: 0.0414
Epoch 36/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0186 - val_loss: 0.0219
Epoch 37/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0210 - val_loss: 0.0329
Epoch 38/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0186 - val_loss: 0.0220
Epoch 39/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0215 - val_loss: 0.0365
Epoch 40/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0175 - val_loss: 0.0242
Epoch 41/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0197 - val_loss: 0.0428
Epoch 42/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0182 - val_loss: 0.0227
Epoch 43/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0222 - val_loss: 0.0344
Epoch 44/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0187 - val_loss: 0.0280
Epoch 45/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0185 - val_loss: 0.0292
Epoch 46/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 0.0174 - val_loss: 0.0274
Epoch 47/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 0.0181 - val_loss: 0.0359
Epoch 48/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 0.0173 - val_loss: 0.0234
Epoch 49/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0195 - val_loss: 0.0339
Epoch 50/50
130/130 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 0.0173 - val_loss: 0.0253
18/18 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step
```

```
📈 Final Evaluation:
MAE : 0.10
RMSE: 0.14
R²  : 0.89
```